

LearnStack: White-Label Curriculum Delivery, as a Service

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LearnStack

White-Label Curriculum Delivery, as a Service

Product Paper — July 2026

1. Executive Summary

LearnStack is a multi-tenant SaaS platform that lets any organization launch a fully branded online academy in a day — their name, their logo, their colors, their domain — and deliver any curriculum through it: video, rich text, audio, documents, embeds, and graded quizzes, with enrollment tracking and verifiable certificates. Delivery is **multi-channel**: full-fidelity web, WhatsApp with real media messages, SMS, and USSD for feature phones — one lesson authored once, delivered everywhere a learner can be reached.

The platform is production-derived, not speculative. It grew out of MNAIR (Make Nigeria AI Ready), a national AI literacy platform that delivered a 7-day, 175-lesson curriculum in five Nigerian languages and ran a live pilot across 254 users in 11 Nigerian states. LearnStack generalizes that engine: where MNAIR delivers one curriculum for one brand, LearnStack delivers any curriculum for any brand, with each customer isolated inside their own tenant.

The business model is straightforward B2B SaaS: training providers, NGOs, government programs, and corporate L&D teams pay a monthly platform fee to run their academy on LearnStack instead of building one or renting a generic LMS that carries someone else's brand.

2. The Problem

Organizations that need to teach at scale face a bad set of options:

1. **Build their own platform.** Six months and a development team most training businesses, NGOs, and agencies do not have.
2. **Rent a hosted course platform** (Teachable, Thinkific, Kajabi). Fast, but the customer's academy lives on someone else's brand and URL structure, pricing scales punitively with students, and content is locked in — there is no portable export of a full curriculum.
3. **Self-host an open-source LMS** (Moodle, Open edX). Free software, expensive reality: server administration, plugin maintenance, and an interface learners tolerate rather than enjoy.

The gap is sharpest for the organizations LearnStack grew up around: African training academies, civic literacy programs, and government initiatives that need **their own brand** in front of learners and funders, need **certificates that third parties can verify**, and need **curricula that move with them** — imported from what they already have, exported when they leave.

3. Evidence: The MNAIR Foundation

LearnStack is the productized core of a platform that already worked:

- **175 lessons** across 12 modules, authored once and delivered in English, Pidgin, Hausa, Yoruba, and Igbo
- **254-user pilot across 11 Nigerian states** with strong lesson-completion engagement
- Government demand signal: NITDA and the NITDA–NYSC digital literacy ambassador program actively seeking platform partners (2026)
- NDPA-compliant data handling (consent tracking, retention discipline)

The pilot proved the delivery engine, the quiz/certificate loop, and the multilingual content model. LearnStack’s bet is that dozens of organizations need exactly this machinery under their own name. The MNAIR curriculum itself imports into LearnStack with one command and serves as both the reference implementation and the first flagship tenant.

4. Product Overview

4.1 WHITE-LABEL MULTI-TENANCY

Every customer is a **tenant** with its own branding: product name, tagline, logo, favicon, primary/secondary colors, footer, and support email. The learner-facing app fetches branding at boot and themes itself at runtime — one codebase, unlimited brands. Tenants resolve per request (`X-Tenant` header or `?tenant=` slug), so mapping a customer’s custom domain to their academy is a one-line reverse-proxy rule. Data isolation is enforced at the API layer on every route and covered by automated tests.

Roles: platform operator (superadmin) → tenant admin → instructor → learner. Tenant admins manage their own branding, staff, and content without touching anyone else’s.

4.2 AUTHOR ANY LESSON FORMAT

Lessons are ordered lists of typed **content blocks**:

Block	What it delivers
<code>text</code>	Rich text / Markdown
<code>video</code>	YouTube, Vimeo, or direct upload (mp4/webm/mov)
<code>audio</code>	Podcast-style lessons, language drills
<code>image</code>	Diagrams, infographics, with captions
<code>file</code>	PDFs, slide decks, worksheets
<code>embed</code>	Any external tool via iframe
<code>quiz</code>	Multiple-choice with pass thresholds, graded server-side

The Studio (admin console) provides a block editor with drag-ordering, media upload, and a live learner preview beside the editor. Answer keys never leave the server; learners get graded results, not the key.

4.3 DELIVER ANY CURRICULUM

The content hierarchy is deliberately simple — **curriculum** → **modules** → **lessons** → **blocks** — because it maps onto everything from a 7-day WhatsApp literacy program to a university short course. Two properties matter:

- **Portability.** A whole curriculum round-trips as one JSON document (`import` / `export` endpoints). Customers arrive with content and can leave with it. A YAML importer already migrates MNAIR-format curricula wholesale.
- **Draft/publish lifecycle.** Curricula are invisible to learners until published; unpublishing is instant.

4.4 MULTI-CHANNEL DELIVERY: WEB, WHATSAPP, SMS, USSD

The same lesson, authored once, reaches learners wherever they are — this is the capability MNAIR proved nationally, now available to every tenant:

- **Web:** full fidelity — embedded/uploaded video, audio, images, files, interactive quizzes.
- **WhatsApp** (Meta Cloud API): lessons arrive as conversation — text messages plus **real video, audio, image, and document messages**, quizzes answered by reply, certificates delivered with their verification code.
- **SMS** (Africa's Talking or Twilio): text chunked to segment limits; media degrades gracefully to a captioned link.

- **USSD** (Africa's Talking `CON` / `END` sessions): text-only delivery for feature phones with no data plan — the same curriculum, no smartphone required.

One conversation engine drives all three messaging channels per tenant: course menu → enroll → lesson delivery → quiz (graded server-side, same engine as the web) → progress → certificate. Each tenant connects its own provider credentials (or platform-shared numbers), keeping the white-label promise on the messaging side. A built-in **conversation simulator** in the Studio lets a tenant test every flow on every channel before connecting a provider account. Drip reminders nudge idle learners (“Reply NEXT to continue”) on a cron cadence — MNAIR’s 7-day program pattern as a feature.

4.5 LEARN, PROVE, VERIFY

Learners self-register inside a tenant, enroll, and progress lesson by lesson. Quizzes gate lesson completion where authors want them to. When a learner finishes every lesson, the platform issues a **certificate with a unique verification code**; anyone — an employer, a funder, a ministry — can check it on a public verification page, no account required. This closes the loop that made MNAIR credible to institutional partners.

5. Architecture and Technology

Layer	Choice	Why
API	FastAPI (Python 3.10+)	Same stack as MNAIR; OpenAPI docs for free
Data	SQLAlchemy 2 — SQLite for dev, PostgreSQL in production	Zero-infrastructure trials, boring-reliable scale path
Auth	JWT + PBKDF2 password hashing	Stateless, standard, no native-dependency headaches
Media	Local disk per tenant, one-function swap to S3/GCS	Ship now, scale later without API changes
Channels	One conversation engine; Meta Cloud API, Africa's Talking, Twilio adapters	Same lesson blocks render per channel capability
Frontend	React + Vite + MUI, runtime-themed per tenant	One deploy serves every brand
Packaging	Docker Compose (Postgres + API + nginx frontend)	Single-command production bring-up

Single database, shared schema, `tenant_id` on every row — the standard SaaS pattern that keeps operations cheap until a customer is big enough to justify dedicated infrastructure (at which point the same image deploys standalone, which is itself a sellable “enterprise isolated” tier).

Test coverage runs the full commercial lifecycle end-to-end: create tenant → brand it → author content → learner enrolls → passes quiz → certificate issued → certificate verifies — plus explicit cross-tenant isolation tests.

6. Target Customers and Use Cases

1. **Training academies and bootcamps** (first paying segment). Tech, trade, and professional trainers across Nigeria and West Africa who currently teach through WhatsApp groups and PDFs. They get a branded academy, structured curricula, and verifiable certificates at a fraction of a custom build.
 2. **Government and NGO programs**. The NITDA–NYSC ambassador model generalizes: each agency or program becomes a tenant with its own identity and reporting, on shared national infrastructure. MNAIR is the working proof.
 3. **Corporate L&D and compliance**. Onboarding and certification for distributed workforces; the certificate verification page doubles as an audit trail.
 4. **Edtech entrepreneurs**. Subject-matter experts who want to sell a course under their own brand without becoming platform engineers — the segment Teachable serves, minus the brand tax and lock-in.
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7. Competitive Landscape

	LearnStack	Teachable / Thinkific	Moodle (self-host)	TalentLMS
True white-label (customer's brand only)	✓ core design	Partial, premium tiers	✓ with effort	Partial
Multi-tenant operator model (run many academies)	✓	✗ one brand per account	✗ one install per org	Limited
Curriculum portability (full JSON in/out)	✓	✗	Partial	✗
Publicly verifiable certificates	✓	Add-on	Plugin	✓
WhatsApp / SMS / USSD curriculum delivery	✓ shipped	✗	✗	✗
Ops burden on customer	None	None	High	None
Low-bandwidth / multilingual heritage	✓ (MNAIR)	✗	Neutral	✗

The defensible position is the combination: **operator-grade multi-tenancy + content portability + certificate verification**, with an origin story and feature bias (multilingual, low-bandwidth, WhatsApp/USSD roadmap) matched to markets the incumbents treat as afterthoughts.

8. Business Model

Per-tenant subscription, priced on active learners rather than content volume (authors should never be punished for creating more):

Tier	Target	Indicative price	Includes
Starter	Solo trainers	\$29/mo (₦ equivalent, local pricing)	1 curriculum live, 100 active learners, LearnStack subdomain
Growth	Academies	\$99/mo	Unlimited curricula, 1,000 learners, custom domain, certificate verification
Institution	Gov/NGO/corporate	\$499+/mo	Unlimited learners, SLA, dedicated onboarding, data-residency options
Sovereign	National programs	Custom	Isolated deployment of the same stack, operator training

Additional revenue levers, in order of build cost: curriculum-import services (migrating a customer's existing content — high-touch, high-margin, already half-automated), certificate volume for institutional verifiers, and, later, a revenue share on paid-course checkout.

Unit economics intuition: the marginal tenant costs storage plus a slice of one Postgres instance — near-zero. The cost center is onboarding, which the Studio, seed tooling, and importer are specifically designed to compress.

9. Roadmap

Now (shipped): multi-tenant white-labeling, block-based authoring with video, curriculum import/export, enrollments, server-side quiz grading, verifiable certificates, MNAIR importer, Docker deployment, lifecycle test suite — **and multi-channel delivery: WhatsApp (with media messages), SMS, and USSD, with a per-channel simulator and drip reminders.** No incumbent LMS ships this; it is the wedge for the African market.

Next 90 days — first revenue:

- Billing integration (Paystack first — Nigerian market — then Stripe)
- S3-compatible media storage and CDN delivery
- Learner analytics per tenant (enrollment funnels, completion, quiz scores, per-channel engagement)
- Custom-domain automation (today: one manual proxy rule per domain)
- Hosted cron for drip cadences (today: tenant-triggered endpoint)

Two quarters — differentiation:

- Cohorts and scheduled program starts (the MNAIR “7-day program” pattern, full version)
- Multi-language curriculum variants as a first-class concept
- WhatsApp interactive buttons/lists for menus and quiz answers
- AI-assisted authoring: draft a lesson’s blocks and quiz from a source document

Four quarters — moat:

- SCORM/xAPI import for corporate migration deals
 - Voice/IVR channel for low-literacy audiences
 - Marketplace: tenants license curricula to each other, LearnStack takes a cut
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10. Risks and Honest Unknowns

- **Willingness to pay** in the first segment is assumed, not yet proven; the 90-day plan exists to get five paying tenants and find out.
 - **Incumbent response:** Thinkific/Teachable could deepen white-labeling. The multi-tenant operator model and channel delivery (WhatsApp/USSD) are the hedges — structural features, not toggles.
 - **Payments friction** in NGN (FX, chargebacks) argues for Paystack-first and annual invoicing for institutions.
 - **Single-database blast radius:** acceptable now, mitigated by the documented per-customer isolated-deployment path.
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11. Summary

LearnStack turns a proven, nationally piloted curriculum-delivery engine into infrastructure any organization can rent under its own name. The product is built and verified end-to-end; the first flagship curriculum (MNAIR) imports in one command; the near-term work is billing, storage, and five design partners — not research.

One line: *Shopify made everyone a store. LearnStack makes everyone an academy.*

Appendix — try it in five minutes: `README.md` covers local setup. Seeded demo: tenant `demo`, admin `admin@demo.local`, sample curriculum plus the full MNAIR English curriculum (7 modules, 35 lessons) pre-imported.